

Proposition The solution of

$$\begin{cases} ax + by = p \\ cx + dy = q \end{cases}$$

is of the form

$$x = \frac{dp - bq}{ad - bc}$$

$$y = \frac{aq - cp}{ad - bc}$$

as long as $ad - bc \neq 0$

*Proofs are required

Proof: logical justification of your claim in words

Take the 1st eq $\times d$ & 2nd eq $\times b$

$$adx + bdy = pd$$

$$bcx + bdy = bq$$

subtract eq:

$$(ad - bc)x = pd - bq$$

$$\frac{ad - bc}{ad - bc} x = \frac{pd - bq}{ad - bc}$$

$$\text{This shows } x = \frac{pd - bq}{ad - bc}$$

Take 1st eq $\times c$ & 2nd eq $\times a$

$$(ax + by = p) c$$

$$acx + bcy = cp$$

$$(cx + dy = q) a$$

$$- acx + day = aq$$

$$\frac{(bc - dc)y = cp - aq}{bc - dc} \quad \text{This shows } y = \frac{cp - aq}{bc - dc}$$

$$y = \frac{pc - aq}{bc - dc} \rightarrow \frac{aq - pc}{ad - bc}$$

Exercise: Try to come up w/ formula

use this notation in lin alg

$$\begin{cases} a_{11}x + a_{12}y + a_{13}z = p_1 \\ a_{21}x + a_{22}y + a_{23}z = p_2 \\ a_{31}x + a_{32}y + a_{33}z = p_3 \end{cases}$$

also this is hard so like alg

section 1.2 Matrices & vectors

ex:

$$\begin{cases} x + 2y + 3z = 14 \\ 3x - y - z = -2 \\ x - y + z = 2 \end{cases}$$

→ convert to matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 14 \\ -2 \\ 2 \end{bmatrix}$$

*forget all the variables just focus on the numbers
Def: An $n \times m$ matrix is simply a collection of nm numbers in rows & cols
matrix

Psychological nothing more

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \xrightarrow[\text{1st} \times 2]{\text{Replace 1st by}} \begin{bmatrix} 2 & 0 & 1 \\ 3 & -1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \xrightarrow[\text{1st} + 2 \cdot 2nd]{\text{replace 2nd by 2nd - 3rd}} \begin{bmatrix} 2 & 0 & 1 \\ 4 & 0 & -2 \\ 1 & -1 & 1 \end{bmatrix}$$

divide by 2

$$\begin{bmatrix} 2 & 0 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ -2 \\ 2 \end{bmatrix}$$

replace 1st by 1st + 2nd

$$\begin{bmatrix} 3 & 0 & 0 \\ 1 & 0 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ -2 \\ 2 \end{bmatrix}$$

notation

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{in} & a_{in2} & \dots & a_{inm} \end{bmatrix}$$

i th row, j th col a_{ij}

1st / 6

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 1 & 0 & -1 & -2 \\ 1 & -1 & 1 & 2 \end{bmatrix}$$

2nd $\leftarrow$ 2nd - 1st

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & -1 & -3 \\ 1 & -1 & 1 & 2 \end{bmatrix}$$

2nd $\leftarrow$ (-1)

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 1 & -1 & 1 & 2 \end{bmatrix} \xrightarrow{\text{3rd} \leftarrow \text{3rd} - \text{1st} - 2nd} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & -1 & 0 & -2 \end{bmatrix}$$

3rd $\leftarrow$ (-1)

$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 1 & 0 & 2 \end{bmatrix} \xrightarrow{\text{swap 2nd \& 3rd}} \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$$